

Adaptive Step-Size Methods and a Polyak Program for Variational Quantum Optimization

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Abstract

We collect the main families of step-size selection for (stochastic) gradient descent, organized by the global quantity each method requires: a smoothness constant, the optimal value, or the distance to the solution. We then focus on the Polyak family and argue that the variational quantum setting is unusually favorable for it, because the optimal value f^* is frequently known or admits a certified lower bound, and because the estimator variance is computable from measurement statistics. We give the quantum objective, a shot-noise-debiased Polyak step, and a list of research directions.

1 Local Lipschitz (L) estimation

A global smoothness constant L yields a single conservative step $\eta = 1/L$, but on a band-limited (trigonometric) landscape the curvature varies sharply, so a step that tracks the *local* Lipschitz constant is preferable. Both methods below estimate curvature from gradients that are computed anyway, with no line search and no prior constants.

Barzilai–Borwein (BB). Approximate the Hessian along the most recent step by a scalar $\eta_k^{-1}I$ and solve the resulting secant condition in least squares. With $\mathbf{s}_k = \boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1}$ and $\mathbf{y}_k = \nabla f(\boldsymbol{\theta}_k) - \nabla f(\boldsymbol{\theta}_{k-1})$,

$$\eta_k^{\text{BB1}} = \frac{\langle \mathbf{s}_k, \mathbf{s}_k \rangle}{\langle \mathbf{s}_k, \mathbf{y}_k \rangle}, \quad \eta_k^{\text{BB2}} = \frac{\langle \mathbf{s}_k, \mathbf{y}_k \rangle}{\langle \mathbf{y}_k, \mathbf{y}_k \rangle}. \quad (1)$$

Each ratio is a finite-difference estimate of $1/L$ along the recent displacement, so the step automatically inflates in flat regions and shrinks in curved ones.

Adaptive gradient descent without descent (AdGD). Estimate the local Lipschitz constant from consecutive gradients and cap the rate of step growth, giving a single-loop method with a rate at least as good as optimally tuned GD (Malitsky–Mishchenko 2020):

$$\eta_k = \min \left\{ \sqrt{1 + \omega_{k-1}} \eta_{k-1}, \frac{\|\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1}\|}{2 \|\nabla f(\boldsymbol{\theta}_k) - \nabla f(\boldsymbol{\theta}_{k-1})\|} \right\}, \quad \omega_k = \frac{\eta_k}{\eta_{k-1}}. \quad (2)$$

The second argument is a local $1/(2\hat{L}_k)$ estimate, while the first lets the step increase between iterations but no faster than $\sqrt{1 + \omega_{k-1}}$.

2 Optimal-value ($f^\star$) and distance (D) estimation

The two adaptive families that give worst-case-*optimal* rates each hinge on one global scalar: the Polyak family needs $f^\star = \min_{\boldsymbol{\theta}} f(\boldsymbol{\theta})$, and the distance family needs $D = \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^\star\|$. Which one is cheaper to obtain decides which family transfers well to a given problem.

Optimal-value estimation. When $f^\star$ is unknown, run the Polyak step against a lower bound $\ell_k \leq f^\star$ and tighten it online toward $f^\star$:

$$\eta_k = \frac{f(\boldsymbol{\theta}_k) - \ell_k}{\|\nabla f(\boldsymbol{\theta}_k)\|^2}, \quad \ell_k \nearrow f^\star. \quad (3)$$

Using any valid lower bound keeps the step well defined and convergent, and schemes such as two-sequence tracking or online estimation drive ℓ_k upward as better function values are observed (Hazan–Kakade 2019; Orvieto et al. 2022).

Distance estimation (D-Adaptation). The optimal step for convex Lipschitz problems is $\propto D/\sqrt{n}$, so estimate D by an increasing lower bound built from accumulated gradients (Defazio–Mishchenko 2023). With $\mathbf{s}_{k+1} = \mathbf{s}_k + \eta_k \mathbf{g}_k$, $\mathbf{s}_0 = \mathbf{0}$,

$$\hat{d}_{k+1} = \frac{\|\mathbf{s}_{k+1}\|^2 - \sum_{i=0}^k \eta_i^2 \|\mathbf{g}_i\|^2}{2 \|\mathbf{s}_{k+1}\|}, \quad \eta_{k+1} = \frac{\hat{d}_{k+1}}{\sqrt{\sum_{i=0}^k \|\mathbf{g}_i\|^2}}. \quad (4)$$

The numerator $\hat{d}_k$ provably lower-bounds D and increases toward it, so the step approaches the optimal $D/\sqrt{n}$ form without ever knowing D ; Prodigy (Mishchenko–Defazio 2024) and DoG (Ivgi et al. 2023) refine the same idea.

3 Polyak descent

The Polyak step chooses, at each iterate, the displacement that would land exactly at the known optimal value if the objective were quadratic along the gradient,

$$\eta_k = \frac{f(\boldsymbol{\theta}_k) - f^\star}{\|\nabla f(\boldsymbol{\theta}_k)\|^2}, \quad \boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta_k \nabla f(\boldsymbol{\theta}_k). \quad (5)$$

It requires knowledge (or a lower bound) of $f^\star$ but no smoothness constant, and it is worst-case optimal for the resulting (sub)gradient method.

The development of this idea spans several variants and analyses:

- **Original Polyak (1963/1987).** Subgradient method for nonsmooth convex minimization; requires $f^\star$; $O(1/\sqrt{k})$ nonsmooth, linear under smoothness and strong convexity.
- **Stochastic Polyak step, SPS / SPS_{max} (Loizou et al. 2021).** Per-sample step $\eta_k = (f_v(\boldsymbol{\theta}_k) - f_v^\star)/(c \|\nabla f_v(\boldsymbol{\theta}_k)\|^2)$ with a cap; linear convergence to the true solution under interpolation, no problem constants.
- **Quasar-convex / arbitrary-sampling analysis (Gower et al. 2020).** Rates for SPS under quasar- (rather than full) convexity, which is the relevant relaxation for nonconvex-but-benign landscapes.

- **DecSPS (Orvieto et al. 2022).** Combines a steady step decrease with SPS adaptivity; converges to the exact minimizer *without* interpolation by removing the step/gradient correlation bias near minima.
- **ALI-G / AdaSPS (Berrada et al. 2020).** Inflate the denominator in the interpolation regime for stability.
- **Moving-target SPS, MOTAPS (Gower–Defazio–Rabbat 2021).** Track a per-point loss as a moving target; drops the interpolation requirement.
- **Momentum variants, MomSPS / MomDecSPS / MomAdaSPS (2024).** Heavy-ball Polyak steps with exact convergence and no interpolation.
- **Accelerated / heavy-ball Polyak (Barré et al. 2020; Goujaud–Taylor–Dieuleveut 2022).** Equivalent to a conjugate-gradient-type method on quadratics, with finite-time convergence and instance optimality.
- **Polyak Minorant Method (Devanathan–Boyd 2024).** Generalizes the subgradient to any convex *minorant* (tight lower bound) and handles constraints; the natural abstraction when a certified lower bound is available.
- **Online f^* estimation (Hazan–Kakade 2019; two-sequence methods 2025).** Maintain and tighten an estimate of f^* when it is unknown.

4 Quantum objective and Polyak integration

Objective. For a parameterized state $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle$ and an observable $H = \sum_j c_j P_j$,

$$f(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle = \sum_j c_j \langle \psi(\boldsymbol{\theta}) | P_j | \psi(\boldsymbol{\theta}) \rangle, \quad \boldsymbol{\theta} \in \mathbb{T}^P, \quad (6)$$

with gradients available exactly in expectation through the parameter-shift rule,

$$\partial_k f(\boldsymbol{\theta}) = \frac{1}{2} \left[f(\boldsymbol{\theta} + \frac{\pi}{2} \mathbf{e}_k) - f(\boldsymbol{\theta} - \frac{\pi}{2} \mathbf{e}_k) \right]. \quad (7)$$

Each evaluation of f and of $\partial_k f$ is an unbiased estimator whose variance is determined by the finite shot count and is therefore computable.

Why Polyak fits. Three properties of this objective make the Polyak family the natural choice. (i) The numerator is directly measured, and f^* is often known ($f \rightarrow 0$ for fidelity/overlap costs) or admits a certified lower bound ($f^* \geq E_0 \geq \underline{E}$ for ground-state energy, via Temple, Lanczos, or semidefinite relaxations). (ii) The denominator follows from parameter shift with a known variance. (iii) Both numerator and denominator are unbiased with *computable* variance, so one can debias rather than merely cap.

Shot-noise-debiased step. Let $\widehat{\|\nabla f\|^2}$ be the plug-in estimate from finite shots and Σ_k the shot covariance of the gradient estimate. The naive denominator is biased,

$$\mathbb{E}[\widehat{\|\nabla f(\boldsymbol{\theta}_k)\|^2}] = \|\nabla f(\boldsymbol{\theta}_k)\|^2 + \text{tr } \Sigma_k, \quad (8)$$

and the reciprocal introduces a second-order bias $\mathbb{E}[1/X] \approx 1/\mathbb{E}[X] + \text{Var}(X)/\mathbb{E}[X]^3$. Subtracting the known shot variance gives the debiased Polyak step

$$\eta_k = \frac{f(\boldsymbol{\theta}_k) - \ell_k}{\|\nabla f(\boldsymbol{\theta}_k)\|^2 - \text{tr } \hat{\Sigma}_k} \quad \ell_k \leq f^*, \quad (9)$$

optionally with the reciprocal-bias correction applied to leading order.

Coupling to shot allocation. Because $\text{Var}(\eta_k)$ is an explicit function of the per-term shot counts N_k , the shots can be treated as a control variable: allocate N_k so that $\text{Var}(\eta_k)$ falls below the threshold required by the convergence analysis, i.e. an iCANS-style controller targeting the *step* estimate rather than the gradient. For variational eigensolvers where the ansatz need not reach E_0 (so f^* is unreachable and unknown), the non-interpolation variants of Section 3 (DecSPS, MOTAPS, MomAdaSPS) port directly, removing any dependence on an interpolation/overparameterization assumption.

5 Discussion

Several further directions follow from the same structure.

Band-limited line search. Restricted to the ray $\boldsymbol{\theta}_k - \eta \nabla f(\boldsymbol{\theta}_k)$, the cost is a finite sum of sinusoids with bounded frequency, so a handful of shifted evaluations determine it and yield a near-exact line search—a quantum-native step rule that is strictly stronger than the generic Polyak step and connects to rotsolve-style coordinate solvers.

Theory backbone: quasar-convexity / PL. SPS rates hold under quasar-convexity rather than full convexity, and a local Polyak–Łojasiewicz inequality upgrades constant-step and Polyak descent to linear convergence. Establishing quasar-convexity or a PL condition for VQA basins (e.g. in the lazy/QNTK regime) would retroactively justify the entire adaptive family here.

Accelerated Polyak as a cheap metric surrogate. In the locally quadratic (lazy) regime, the conjugate-gradient equivalence of heavy-ball Polyak steps provides near-optimal stepping that captures curvature without forming the quantum Fisher information, offering a cheaper stand-in for quantum natural gradient.

Distance estimation on the torus. The D-Adaptation/Prodigy family estimates $D = \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|$, which on the periodic parameter torus $\mathbb{T}^P$ has a different geometry than in $\mathbb{R}^P$; characterizing this could yield a complementary parameter-free method when no lower bound on f^* is available.